

AI Evaluation Methodology for Use Cases Requiring Complete Confidentiality

Version 2 · reviewed against the A-IEP pipeline source and the companion docs in [docs/](#)

Prepared 6 August 2026 · Anchors quote V.2 verbatim · Paste-ready replacement text is boxed in green

How this review is organised. V.2 already fixed several things flagged against V.1: the redaction claim now admits that names and dates are retained, the OpenAI enterprise no-training term is stated, the truncated sentence is gone, the contradictory "True or False?" description of the verifier has been removed, and the odd "(V2)" notation on GEMBA-MQM is dropped. Those are not repeated below.

Reading `EVAL_FEASIBILITY_COST.md` and `BLOG_SYNTHETIC_IEP_DATASET.md` also changed the picture. A large share of what looked like gaps in the methodology doc turns out to be already decided: cost, cadence, cross-family judging, citation validation, the deterministic layer, human calibration, and a deliberately messy 50-case corpus. Those are not gaps in the programme. They are gaps in the *publishable artifact*, which is the only thing an outside reader will see. Part 2 covers them separately for that reason.

Part 1 corrections, where V.2 states something the code contradicts. **Part 2** decisions already made elsewhere but invisible in this document. **Part 3** genuinely open in every document. **Part 4** references and line edits. **Part 5** what the recommendations cost, priced as a delta at $n = 100$ documents per month. **Part 6** things only you can verify.

STATUS AT A GLANCE

ID	ITEM	WHERE IN V.2	PRIORITY
C1	Translation compares against the English summary, not the IEP	Translation Quality	P1
C2	The 8th-grade reading level claim is backwards	Quality of Summary	P1
C3	Backfilled sections ship a false page-1 citation	Factual accuracy	P1
C4	Schema-failure claim is stronger than the code	Factual accuracy	P2
C5	"only keeps the names and dates" reads backwards	Our Case Study	P2
C6	Three retrieval tools, actually four	Factual accuracy	P3
C7	Arabic is live outside production, not "soon"	Translation Quality	P3
C8	"the system" should name the vendor	Our Case Study	P3
D1	Human calibration plan is invisible here	throughout	P1
D2	Cross-family judge never stated	throughout	P1
D3	Deterministic layer absent	Quality, Translation	P2
D4	Citation validation still a dropped sentence	Factual accuracy	P2
D5	No operating model: cost, cadence, gate	missing	P2
D6	Real IEPs in calibration need pre-empting	Sources of Truth	P2

ID	ITEM	WHERE IN V.2	PRIORITY
G1	The redaction step is never evaluated	missing	P1
G2	No stage attribution; OCR unmeasured	missing	P1
G3	No threat model	Guiding Question	P2
G4	Corpus representativeness asserted, not measured	Sources of Truth	P2
G5	The fact ledger is circular	Sources of Truth	P2
G6	Warmth rubric rewards softening bad news	Quality of Summary	P2
G7	IEP facts do not decompose flatly	Factual accuracy	P2
G8	Reading target: 5th-7th vs 8th grade	Quality of Summary	P3
G9	No subgroup slicing	missing	P3
G10	Four hard cases missing from the corpus plan	Sources of Truth	P3
G11	No scope and limitations statement	missing	P3
G12	The novel contribution is buried	Philosophy	P2

PART 1 · CORRECTIONS

Places where V.2 asserts something the A-IEP source code contradicts. Each was checked against the file cited.

C1 Translation is measured against the wrong source **P1**

ANCHOR

"Essentially this is a process that provides specific severity determinations and error types in response to finding differences as compared to the source document and returning a JSON object output."

ISSUE

A-IEP does not translate the IEP `translate_content/handler.py` fetches the stored English parsing result and translates that. The translation agent never sees the source document. So "as compared to the source document" describes a comparison the system cannot make, and it conflates two very different measurements. The consequence is not cosmetic: because translation is a pivot off the summary, a summarization hallucination is translated faithfully into four languages and scores clean under MQM. Reporting only the MQM number produces a green translation dashboard sitting on top of a broken summary.

CHANGE · INSERT AT THE TOP OF "WHAT NEEDS TO BE MEASURED?"

One structural fact drives this whole section: A-IEP does not translate the IEP. It translates the English summary the parsing step produced, and the source document is never seen by the translation agent. That means there are two distinct things to measure and they are easy to conflate.

Translation fidelity is the English summary compared against its translation. This is what GEMBA-MQM measures well and is the right target for error typing and severity.

End-to-end accuracy is the translated summary compared against the original document. This is what the parent actually receives, and it is not what MQM measures. Because translation is a pivot off the summary, an error introduced during summarization is translated faithfully into every language and scores clean under MQM.

We therefore run both: GEMBA-MQM against the English summary for translation quality, and a second pass of the atomic-fact pipeline on the translated text against the source document's fact ledger for end-to-end accuracy. Where the two diverge, the divergence localises the error to the summarization step or the translation step, which is the stage-attribution principle applied to language.

C2 The 8th-grade reading level claim is backwards P1

ANCHOR

"All text is prompted to use an 8th grade reading level, even when translated."

ISSUE

The reverse is true. The 8th-grade instruction appears only in the four translation prompts in `translate_content/config.py`. Grepping the English parsing prompt for "grade" returns zero hits: English has no reading-level instruction at all. The source text of every translation is therefore unconstrained on exactly the dimension this section is about.

CHANGE · REPLACE THE PARAGRAPH

Each summary is prompted to be "warm, supportive, and student-specific" and to "use friendly, encouraging language, and aim for a tone that is informative yet comforting to families and educators who read it." A reading level is specified in the translation prompts, which request an 8th-grade level in each target language. The English prompt currently specifies no reading level at all, which means the source text of every translation is unconstrained on the dimension we most want to control. Exposing findings like this is what the baseline is for.

C3 Backfilled sections ship a false citation P1

ANCHOR

"a code-level backfill inserts the 'not found' placeholder for any section the model dropped, so an omission cannot quietly become an invention."

ISSUE

True as far as it goes, but `_ensure_complete_english_sections` also hardcodes `'page_numbers': [1]` on the inserted section, and that citation is rendered to the parent. It is a hallucinated citation produced by our own code, not by the model. The feasibility doc tracks placeholder-section rate but not this. Since the paragraph is arguing that the backfill is a safety property, the caveat belongs in it.

CHANGE · APPEND TO THE SENTENCE

One caveat. When the backfill inserts a placeholder it also inserts a citation to page 1, which is not a real citation. This is a defect our own code produces, it is displayed to the parent, and the citation-validation metric described below detects it.

C4 The schema-failure claim is stronger than the code P2

ANCHOR

"If the output fails schema validation, the document fails the pipeline rather than shipping a partial summary."

ISSUE

There is a recovery path. On a model behaviour error the code salvages `e.final_output` and re-validates it before giving up (`open_ai_agent.py:137-145`). Also, what is guaranteed is a complete *section set*, not a complete summary.

CHANGE · REPLACE THE SENTENCE

If the output cannot be validated, the document fails the pipeline rather than shipping an incomplete set of sections. There is one recovery path: when the agent framework raises a model behaviour error, the code attempts to salvage a partial output and re-validate it, failing only if that also does not conform.

C5 "only keeps the names and dates" reads backwards P2

ANCHOR

"creating a redacted markdown version of each document which only keeps the names and dates in the document"

ISSUE

The substance is now right, but the phrasing inverts on first read: "only keeps the names and dates" scans as "discards everything except names and dates", which would describe a document with no content left. It also still implies de-identification, which this is not. Worth being precise, because this sentence carries the paper's central confidentiality claim.

CHANGE · REPLACE THE TWO SENTENCES

A-IEP limits what leaves the system by redacting the OCR text before any of it reaches the summarization model. AWS Comprehend detects PII on each page and the pipeline replaces every detected entity with a type label, with two deliberate exceptions: personal names and dates are retained, because a summary that cannot name the student or the dates of their meetings is not useful to a parent. The redacted markdown is therefore not de-identified. It is stripped of contact details and identifier numbers but still contains names, dates, and a description of the child's disability. What redaction buys is a reduction in incidental identifiers, not anonymity. The confidentiality guarantee rests on access control: the redacted text goes to OpenAI under an enterprise agreement that excludes it from training, the summary returns to the uploading parent alone, and no member of the development or evaluation team can read either.

C6 Three retrieval tools, actually four P3

ANCHOR

"exposed through three retrieval tools that return the full text or specific pages"

ISSUE

Accurate about the three content tools, but there is a fourth, `get_section_info`, which serves `SECTION_KEY_POINTS`. That matters for evaluation, not just description: those key-point lists are already the per-section checklist your coverage metric proposes to build. Saying so closes the loop between the two halves of the paper.

CHANGE

exposed through four tools: three that return document content (the full text, a single page, or a set of pages) and one that returns the curated key points for a given section. That fourth tool matters for evaluation as well as generation, because the key-point lists it serves are already a per-section checklist of what should be present. The completeness metric below reuses them rather than inventing a new checklist.

C7 Arabic status P3

ANCHOR

"Currently A-IEP offers translation in Spanish, Vietnamese, Mandarin, and Arabic soon."

CHANGE

Currently A-IEP offers translation in Spanish, Vietnamese, and Mandarin in all environments, and Arabic, which is enabled outside production and pending release there.

C8 "the system" should name the vendor **P3**

ANCHOR

"The final summarizations are only seen by the system and the end-user who uploaded that IEP"

ISSUE

"The system" is doing a lot of quiet work here. You state the OpenAI enterprise term two sections later, which is the right fact in the wrong place: a sceptical reader forms their view of the confidentiality claim in the case study and does not revise it afterwards. Move it forward.

CHANGE

Within the application, the final summaries are visible only to the parent who uploaded that IEP. They are also processed by OpenAI under an enterprise agreement that excludes our data from training. No member of the development or evaluation team can read them.

PART 2 · ALREADY DECIDED, BUT INVISIBLE IN THIS DOCUMENT

Each of these is settled in `EVAL_FEASIBILITY_COST.md` and absent from the methodology doc, which is the artifact that gets published and read. The fix is a short paragraph folding the decision in, or an explicit cross-reference. As it stands the methodology reads as less rigorous than the programme actually is, which is the wrong way round.

D1 The human calibration plan is the answer to the obvious objection, and it is missing **P1**

ISSUE

As written, V2 is entirely LLMs judging LLMs. FActScore and G-Eval both established their validity by reporting agreement with human annotators, and a reader who knows that will notice you adopted the methods and dropped the validation. You have not dropped it: the feasibility doc commits to quarterly human MQM-lite reviews per language, calibration against four or five real IEPs with partner verification, and report-only status until variance is baselined. That is a good answer, and nobody reading the methodology doc can see it.

CHANGE · NEW SUBSECTION AT THE END OF "THE PHILOSOPHY BEHIND THE METHODOLOGY"

Human alignment

Every method below uses a language model as a judge. FActScore and G-Eval both established their validity by reporting agreement with human annotators, and adopting a method without that step leaves us with numbers that cannot be shown to be wrong. No metric in this document is treated as a baseline until it has been calibrated against human judgement, and every metric is report-only until its run-to-run variance has been measured.

Calibration has three parts. Annotators score a sample of summaries generated from synthetic documents, and we report both inter-annotator agreement and judge-to-human agreement. Translation adds a periodic human review per language expressed in MQM vocabulary, so that judge output and human output are directly comparable. And a small set of real IEPs, contributed with explicit consent by partners who verify the results themselves, anchors what the scores mean on documents we did not write.

D2 The cross-family judge is never stated **P1**

ISSUE

The doc raises inter-family bias for the synthetic *generator* and never raises it for the *judge*, which is where it bites harder: GPT-4.1 writes the summary, and if a GPT model grades it you inherit self-enhancement bias on every headline number. You have already decided against this, with Claude on Bedrock judging GPT-4.1 output and the added benefit of keeping judge traffic inside the AWS account. Say so where you introduce G-Eval and the atomic-fact verifier.

CHANGE · ADD TO "HOW TO EVALUATE FACTUAL ACCURACY?"

Judging is deliberately cross-family. The summarization model is GPT-4.1, and using a judge from the same family would invite self-enhancement bias on every headline number, the same concern we raise about the synthetic generator. Judges are Claude models accessed through AWS Bedrock, which also keeps judge traffic inside the same account and under the same agreements as the rest of the pipeline. For metrics that gate a release we sample a second judge periodically and report the agreement between them, so that judge drift is visible rather than assumed away.

D3 The deterministic layer is the cheapest and most defensible part, and it is absent **P2**

ISSUE

Placeholder-section rate, glossary adherence against the per-language dictionaries, numeral and date preservation including the Arabic Western-digit rule, structure preservation, English Flesch-Kincaid, and per-stage timing metrics. Zero model cost, no judge, no alignment problem, and shipped first. For a paper whose thesis is evaluation under confidentiality, a layer of metrics that needs no judge and reveals no content is directly on-theme, and it is currently not mentioned once.

CHANGE · NEW SUBSECTION, REFERENCED AGAIN FROM THE TRANSLATION SECTION

The deterministic layer

Before any judged metric, a layer of checks runs in code with no model involved. Placeholder rate records how often a section was backfilled as "not found", which is currently invisible. Glossary adherence checks, for every approved term present in the English summary, whether its approved rendering appears in the translation, using the per-language dictionaries our translators already built. Numeral and date preservation checks that quantities survive translation, including the Arabic rule requiring Western digits. Structure preservation checks that section and list structure survives. Flesch-Kincaid records the English reading grade. Per-stage timing and token counts record what the pipeline did.

These cost nothing, they are reproducible, they cannot disagree with themselves between runs, and none of them requires a human or a model to read the document. They ship first and they form the floor that the judged metrics sit on top of.

D4 Citation validation is still one dropped sentence P2

ANCHOR

"Confirming the accuracy of our citations can be folded into the AFG process."

ISSUE

The feasibility doc treats this properly, with a genuinely elegant design: validate each atomic fact against its cited page only, so a fact that validates against the document but not against its cited page is a citation error, at no extra cost. The methodology doc gives it eleven words and moves on. It deserves promotion, because the citation is the only part of the output a parent can verify independently, which makes it the mechanism that substitutes for the developer inspection your premise forbids. It is the most on-thesis metric you have.

CHANGE · REPLACE THE SENTENCE WITH A SUBSECTION

Citation accuracy

Each section returns the page numbers it was drawn from, and those numbers are shown to the parent. This is the only part of the output a parent can check independently, which makes it the mechanism that stands in for the developer inspection we cannot perform, and it deserves its own metric rather than a footnote.

It comes almost free from the atomic-fact design. Each fact is validated against its cited page rather than against the whole document, so a fact that validates against the document but not against the page it cites is precisely a citation error. Citation precision is the proportion of cited pages that actually contain content belonging to that section; citation recall is the proportion of pages containing that section's content that the summary cites. Deterministic sanity checks run alongside: a cited page beyond the document's page count, or a placeholder section citing page 1, are detectable in code with no model involved.

Together these catch a failure that groundedness alone does not. A claim that is true of the document but attributed to the wrong page sends a parent to the wrong place in a fifty-page record, and it does so while every factuality metric reports success.

D5 No operating model: cost, cadence, gate, reproducibility **P2**

ISSUE

A methodology with no cost, cadence or decision rule reads as aspirational, and "record a baseline, improve, re-evaluate" is a loop with no exit condition. You already have the numbers: roughly \$0.25 to \$0.55 per document, \$25 to \$110 per month at current volume, \$30 to \$70 for a full golden run, and a four-phase build order. A short section converts the paper from a proposal into a plan.

CHANGE · NEW SECTION BEFORE REFERENCES

Operating the evaluation

A methodology without a cost, a cadence and a decision rule tends to be run once and abandoned.

Cost. Background scoring of a document across all metrics runs at roughly \$0.25 to \$0.55, or \$25 to \$110 per month at current volume. A full offline run over the golden corpus is \$30 to \$70. All evaluation is asynchronous and admin-side, so parent-facing latency is untouched.

Cadence and gate. The deterministic layer runs on every document. Judged metrics run as a background pass per document, and as a full run over the golden corpus before each promotion to production. Deploys are already gated on the test suite, and the evaluation is only real once a regression on a named metric can block a promotion the same way. Thresholds are set after the baseline exists, not before, and every metric is report-only until its variance is known.

Reproducibility. Judge model versions are pinned and recorded with each result, and reported numbers on golden runs are the median of three samples. A baseline that cannot be re-run after a model deprecation is not a baseline, so corpus, prompts, judge versions and harness are versioned together.

D6 Real IEPs in the calibration set need pre-empting **P2**

ISSUE

The plan calibrates against four or five real IEPs, and the feasibility doc lists consent and provenance as an open item. A reader of a paper titled *Requiring Complete Confidentiality* will treat "we used real IEPs" as a contradiction of the premise unless you get there first. Handled well it is a strength, because it shows the boundary is about uncontrolled access rather than a blanket prohibition, and that partners verified results on their own documents rather than us reading them.

CHANGE · ADD TO "OUR SOURCES OF TRUTH"

One deliberate exception to the synthetic-only rule. A small number of real IEPs, contributed with explicit informed consent by partner families or organisations, anchor what the metrics mean on documents we did not author. The evaluation team does not read them. Partners run the tool on their own document and verify the output themselves, and only their judgement, not the document or the summary, returns to us. This is the narrowest form of real-world grounding compatible with the constraint, and it is the only point in this methodology where a real record is involved at all.

PART 3 · GENUINELY OPEN

Not addressed in the methodology doc, the feasibility doc, or the dataset plan.

G1

The redaction step is never evaluated

P1

ISSUE

This is the largest gap in the whole programme. Redaction is the precondition for every confidentiality claim in the paper, and no document proposes to measure it. Comprehend's recall on IEP-shaped text is an empirical question with a direct privacy consequence, and synthetic documents are the ideal instrument, because you plant the identifiers yourself and therefore know the ground truth exactly.

For a paper whose premise is confidentiality, this is also the most defensible number available: it is the number that establishes the premise rather than assuming it. Note that V.1 had a sentence gesturing at this which was truncated mid-clause; V.2 removed the sentence rather than completing it, so the gap is now silent as well as open.

CHANGE · NEW SECTION AFTER "THE PHILOSOPHY BEHIND THE METHODOLOGY"

Evaluating the redaction step

Redaction is the precondition for every confidentiality claim in this document, and it is currently unmeasured. Synthetic documents make it measurable, because we plant the identifiers ourselves. Each generated document carries a ledger of planted PII: street addresses, phone numbers, email addresses, identifier numbers, sibling and staff names, school and district names. We run the real redactor over the real OCR output and report recall per entity type, meaning the proportion of planted identifiers of each type that were detected and replaced. We report false positives too, because over-redaction destroys facts and presents downstream as a summarization failure.

Two limits belong in the report. Names and dates are retained by design, so they are excluded from the recall calculation and their retention is stated rather than scored. And the redactor is invoked with an English language code, so a document containing substantial non-English text sits outside the measured envelope until we test it.

G2 No stage attribution, and OCR is unmeasured **P1**

ISSUE

Every metric in the paper is computed on the last stage's output, so every upstream failure presents as a summarization failure. If OCR drops a services table the coverage metric blames the summarizer; if redaction over-redacts a date range, the same thing happens. This matters more here than in most pipelines, because the corpus plan deliberately introduces OCR noise, duplicate scanned pages and illegible handwriting. You are engineering exactly the conditions under which misattribution occurs, without a way to detect it.

CHANGE · NEW SECTION, IMMEDIATELY AFTER THE REDACTION SECTION

Stage attribution

A-IEP is a pipeline: OCR, redaction, summarization, translation. Measuring only the final output means every upstream failure presents as a summarization failure. If OCR drops a services table, the summary is missing a fact and the coverage metric blames the summarizer. If redaction removes a date range, the same thing happens.

Because synthetic documents have a known ground truth text, we can cut the pipeline at each boundary and measure the transition: OCR output against the known document text, redacted output against OCR output, summary against redacted output. An error is then attributed to the stage that introduced it, and summarization metrics are reported both conditioned on clean OCR and unconditioned. This matters for our corpus specifically, because we deliberately introduce scan noise, duplicate pages and handwriting. Without stage attribution we would be manufacturing the exact conditions that cause misattribution and then reading the result as a model failure.

G3 No threat model **P2**

ISSUE

"Developers and evaluators should not view any related content" treats confidentiality as one undifferentiated boundary, and your own system has at least five: our team, AWS, OpenAI, CI artifacts, CloudWatch. The paper's contribution depends entirely on which of these is being excluded. You now state the OpenAI term, which is most of the raw material; it needs to be a frame at the top rather than a detail in the middle.

CHANGE · NEW SUBSECTION DIRECTLY AFTER THE GUIDING QUESTION

Threat model

"Confidential" is not one boundary, and this methodology only makes sense against a specific one. A-IEP document content is visible to the model vendor, which receives redacted OCR text under an enterprise agreement excluding it from training; to the cloud provider, which stores it under a business associate agreement; and to the parent who uploaded it. It is not visible to the A-IEP development team, the evaluation team, CI artifacts, or anyone with log access, all of which are prevented by access control and by a logging discipline that forbids writing document content, OCR text or summaries to logs.

The constraint this methodology addresses is the second list. We cannot read the inputs or the outputs of the system we are responsible for, and every technique below is chosen for whether it works without a developer ever seeing a real document.

G4 Corpus representativeness is asserted and then abandoned **P2**

ANCHOR

"While IEP creation guides exist the quality and diversity of actual IEPs is difficult to recreate."

ISSUE

You raise the objection and then proceed without answering it. There is an answer available, and it is one of the genuinely novel things this paper could contribute: the deterministic layer you are already building emits per-document statistics that contain no content at all, so they can be computed over real production traffic and compared against the same statistics over the synthetic corpus. That converts "we hope this is representative" into a measured claim, and it names the direction in which the corpus is wrong.

CHANGE · NEW SUBSECTION IN "OUR SOURCES OF TRUTH"

Representativeness we can measure without reading anything

The obvious objection to a synthetic corpus is that we cannot know whether it resembles the documents the system actually receives, precisely because we cannot look at those documents. We can, however, compute statistics over them that carry no content. The deterministic layer already emits, per document, page count, which of the nine sections were found versus reported absent, placeholder rate, OCR confidence, the histogram of PII entity types detected, summary and section lengths, and glossary term hit counts. None of these values contain text from the document.

Comparing their distribution against the same statistics over the synthetic corpus turns representativeness from an assumption into a measurement, and it names where the corpus is thin. If real documents average forty pages and ours average twelve, or if real documents report a section absent five times as often as ours do, we know which direction to fix without anyone having read a single record.

G5 The fact ledger is circular **P2**

ANCHOR

"To create the synthetic data first a ledger of facts will be generated which is given to an LLM to fill out the provided templates of IEPs."

ISSUE

The ledger is both the generation input and the evaluation gold standard. The same artifact that tells the model what document to write also decides whether the summary covered it, so an incomplete or incorrect ledger produces metrics that cannot detect their own error. Worth naming, because a reviewer will.

CHANGE · APPEND TO THAT PARAGRAPH

One risk to state plainly: the ledger is both the generation input and the evaluation gold standard. The same artifact that tells the model what document to write also decides whether the summary covered it, which means an incomplete or incorrect ledger produces metrics that cannot detect their own error. For a sample of documents the ledger is therefore audited by someone who did not author it, against the rendered document rather than against the generation prompt.

G6 The warmth rubric would reward the failure mode **P2**

ANCHOR

"An example of one of the rubrics could be for assessing tone where the 1 is cold and bureaucratic and the 5 is warm and supportive."

ISSUE

The prompt asks for comforting language and the news in an IEP is frequently bad: a service reduced, a request denied, a goal not met. A rubric that scores warmth alone gives its highest mark to a summary that softened a denial of service into ambiguity. There is a second, related axis with nothing measuring it: a summary can be perfectly grounded and still leave a parent worse off by dropping a consent deadline or omitting that minutes were cut from last year. Given that the stated purpose of A-IEP is comprehension *and* advocacy, that omission is conspicuous.

CHANGE · REPLACE THE EXAMPLE SENTENCE

Two axes here need care that a generic summarization rubric would miss.

First, tone is in direct tension with fidelity. The prompt asks for warm and comforting language, and the news in an IEP is often not good: a service reduced, a request denied, a goal not met. A rubric scoring only warmth would give its highest mark to precisely the failure we most want to avoid, which is bad news softened into ambiguity. We therefore score the tension rather than the warmth, penalising a summary for omitting or softening an adverse finding regardless of how supportive it reads.

Second, and specific to A-IEP's purpose, a summary can be entirely grounded and still leave a parent worse off: dropping a consent deadline, omitting that minutes were reduced from the previous year, or failing to surface that a parent may disagree with a placement decision. Because the tool exists to support advocacy and not only comprehension, we add an axis for preservation of actionable rights and deadlines, scored against the rights and time-bound obligations planted in each synthetic document.

G7 IEP facts do not decompose flatly **P2**

ISSUE

FactScore was validated on biographical text, where facts are largely flat and independent. IEP content is heavily conditional, and naive decomposition either drops the qualifiers or splits them into fragments that appear to contradict each other. Since atomic facts are the workhorse of the entire design, the assumption that the domain supports them should be stated and tested rather than inherited from the paper.

CHANGE · ADD TO "HOW TO EVALUATE FACTUAL ACCURACY?"

One caveat specific to this domain. FactScore was validated on biographical text, where facts are largely flat and independent. IEP content is heavily conditional: "thirty minutes per week of speech therapy in a small group setting, to be reviewed at the annual IEP" is one claim with three qualifiers, and naive decomposition either drops the qualifiers or splits them into fragments that appear to contradict each other. Our decomposition prompt keeps qualifiers attached to the claim they modify, and the human calibration round deliberately oversamples conditional and compound facts to confirm that it does.

G8 Reading target contradicts the system **P3**

ANCHOR

"Using an LLM to evaluate readability to a 5th to 7th grade range is a good standard for public health related content."

ISSUE

Your prompts specify 8th grade. Carrying both numbers in one document without acknowledging the difference reads as an oversight.

CHANGE · APPEND

Our prompts currently specify 8th grade. We should choose one target deliberately and record the reason rather than carrying two numbers in the same document. Whichever we choose, the prompt-based difficulty judgement is the measurement, with Flesch-Kincaid reported alongside it as a cheap and reproducible sanity check rather than as the criterion.

G9 No subgroup slicing **P3**

ISSUE

The corpus already varies eight state templates, ages from preschool to transition, meeting types, placements and family languages. Nothing says the metrics will be reported along those axes, and a single average is exactly where a failure concentrated in one population disappears. The slicing is nearly free given the corpus design.

CHANGE · ADD TO "HOW TO EVALUATE SUMMARY QUALITY"

All metrics are reported sliced by the axes the corpus already varies, meaning template profile, age band, placement type and document length, as well as in aggregate. The generator controls each of these, so the slicing costs nothing to produce, and a single average is precisely where a failure concentrated in one population would disappear.

G10 Four hard cases missing from the corpus plan **P3**

ISSUE

The dataset plan already covers missing sections, split tables, duplicate scanned pages, illegible handwriting and consecutive-year pairs, which is a strong list. Four cases are absent, and each targets a failure the current metrics would not otherwise catch.

CHANGE · ADD TO THE CORPUS DESCRIPTION

- Pages from two different students bound into one PDF, which tests whether the summary silently merges two children.
- Contradictory dates between the cover page and the body, which tests whether the model reconciles, picks, or invents.
- An amendment or addendum superseding an earlier goal, which tests whether the summary reports the current state or the original.
- Service minutes given in an ambiguous unit, such as per week versus per month, which is the highest-consequence quantitative error the tool can make.

The absent-section case needs one refinement: because the pipeline backfills a placeholder for any dropped section, the evaluation must distinguish "the model reported not found" from "the code inserted not found". Without that distinction the backfill makes the model look correct by construction.

G11 No scope and limitations statement **P3**

CHANGE · NEW SECTION BEFORE REFERENCES

Scope and limitations of version 1

This version covers summarization and translation. It does not cover text-to-speech output quality or the user interface. It measures the redaction step's recall but does not attempt a formal re-identification risk assessment. It relies on synthetic documents for ground truth, with the representativeness of that corpus measured indirectly rather than established directly, and on a small consented set of real IEPs for anchoring. Beyond aggregate content-free statistics it produces no evidence about the real document population. The strongest claim it supports is that the system behaves in a given way on documents we constructed to resemble real ones, and that those documents resemble the real ones along every dimension we are able to measure without reading them.

G12 The novel contribution is buried **P2**

ISSUE

Most of the document's length goes to FActScore, G-Eval and MQM, which any reader can look up, and comparatively little to the actual constraint in the title. Someone reading for the confidentiality angle finds a competent survey of standard evaluation methods with the interesting parts thinnest. Once G1 through G4 are in, you have four contributions that are genuinely yours, and they should be announced.

CHANGE · NEW SUBSECTION IN "THE PHILOSOPHY BEHIND THE METHODOLOGY"

What is novel here

Atomic Facts, G-Eval and MQM are established methods and we adopt them as published. The contribution of this document is not those methods but what a confidentiality constraint forces around them:

- A synthetic corpus whose representativeness is measured rather than assumed, by comparison against non-content distributional statistics computed over real production traffic.
- Content-free measurement as a first principle, in which an entire layer of metrics runs in code, emits only numbers, and requires neither a judge nor a reader.
- Citation checkability treated as a first-class metric, because the page citation is the only part of the output a parent can verify independently, and it is what substitutes for the developer inspection we cannot perform.
- Direct measurement of the de-identification step, since that step is the precondition for every other claim in this document.

The five areas we address follow from this: de-identification, stage attribution, hallucination and factual accuracy, quality of summary, and translation. The first two are absent from most evaluation literature because most evaluators can read their data.

PART 4 · REFERENCES AND LINE EDITS

References

- **Drop or replace two sources.** "Towards Hallucination-Resilient AI" (ResearchGate) has no author, no date and no venue. "A Concise Review of Hallucinations in LLMs" is cited with a bare arXiv ID and no authors. Neither is citable in a methodology paper that expects scrutiny.
- **FActScore is listed twice.** Delete the bare-ID entry and keep Min et al. (2023).
- **Add missing citations** for GEMBA-MQM (Kocmi and Federmann, 2023), COMET-Kiwi (Rei et al., 2022), and Flesch-Kincaid. All three are discussed substantively and none is cited.
- **Chataigner** is cited in-text as both "(Chataigner)" and "(Chataigner 2024)". Standardise, and move the entry into alphabetical position.

Line edits

- "while thehe MQM methodology" to "while the MQM methodology". New typo in V2.
- "Flesh-Kinkaid" to "Flesch-Kincaid". Appears twice, and the correct spelling is already in the Rooein reference title on the same page.
- "AI-EP summarizes real student IEPs" to "A-IEP". Last remaining spelling variant; the rest of V2 is now consistent.
- "our team at AI for Impact" versus the Burnes Center framing elsewhere. Pick one as the authoring body and use the other only for the GrantWell attribution.
- Heading "How we Currently Evaluate Summary Quality" describes prompting, not evaluation. Retitle to "How A-IEP currently constrains summary quality", matching the parallel heading in the factual accuracy section.
- "An existing issue in the FActScore framework is that if facts are given in an order that misconstrues the meaning there is no built-in way to amend this." This sentence does not land. Suggested: "A known limitation of FActScore is that it scores facts independently and so cannot detect meaning distorted by the order in which facts are presented. We record this as a limitation; the coherence rubric under G-Eval is a partial and indirect check."
- "To use some of these evaluation methods such as RAGAS, we need synthetic data to test against." RAGAS appears once and is never used again, and A-IEP has no retrieval corpus, since the whole document is the context. Either cut it, or note that only the faithfulness formulation carries over.
- "While much of this hinges on having real data some pieces of it can be employed to ensure our dataset maintains enough diversity." Suggested: "SynthTextEval's diversity analysis assumes a real reference corpus, which we do not have. Its intra-corpus measures still apply to the synthetic set on its own, and we use them to detect the failure mode where a generator produces fifty near-identical documents."

PART 5 · WHAT THESE RECOMMENDATIONS COST

Everything in Parts 2 and 3 is an addition to a plan that is already costed. This section prices only the *delta*, so it can be judged on its own. The headline: at 100 documents per month the recommendations add roughly **\$1 to \$3 per month** in recurring API spend and **\$8 to \$34 per full golden run**, against a baseline of \$25 to \$55 recurring. The real cost is not dollars. It is four to six engineering days and roughly 37 person-hours of human review per calibration cycle.

Assumptions

Inherited from `EVAL_FEASIBILITY_COST.md`, unchanged: average document is ~15k tokens of redacted OCR; English output (summary plus nine sections) is ~4k tokens; each translation is ~4k tokens; ~60 atomic facts per document; four active translation languages; GPT-4.1 at \$2/\$8 per million tokens in/out, GPT-4.1-mini at \$0.40/\$1.60, Claude Sonnet on Bedrock at \$3/\$15, Claude Haiku at \$1/\$5. All evaluation is asynchronous and admin-side, so none of it touches parent-facing latency.

Set for this analysis: `n = 100 documents per month`, the low end of the 100 to 200 range in the feasibility doc, chosen so the numbers are a floor rather than a best case. Golden corpus is 54 cases, the planned 50 plus the four hard cases in G10. Full golden runs happen weekly.

My own, and the ones most worth challenging:

- AWS Comprehend `DetectPiiEntities` bills per 100-character unit at \$0.0001, so a ~60k-character document costs about \$0.06 to re-scan. *Verify at implementation*, in keeping with the price caveat already in the feasibility doc.
- The redaction recall test and stage attribution run **offline only**. Both require known ground truth, which exists only for synthetic documents, so neither can run in production and neither adds per-document cost.
- The cross-lingual end-to-end pass runs **on the golden corpus only, not per production document**. This is a judgement call and it is where most of the potential cost sits. Rationale below.
- Scoring redaction recall and OCR fidelity is pure code (span comparison, character error rate) with no model in the loop, so those lines are \$0 beyond the Comprehend call itself.

Recurring, per production document

ITEM	SOURCE	\$/DOC	\$/MONTH AT N=100
Existing plan, all metrics	feasibility doc	0.25 - 0.55	\$25 - \$55
Tone-tension and rights/deadlines rubrics (G6)	added	0.01 - 0.03	\$1 - \$3
Subgroup slicing (G9)	added	0	\$0
Representativeness statistics (G4)	added	0	\$0
Conditional-fact decomposition change (G7)	added	0	\$0
Total		0.26 - 0.58	\$26 - \$58

G6 is the only recurring adder and it is small because the two new axes attach to a G-Eval call that already has the summary in context. The marginal cost is extra rubric text in and two more scores out, roughly 20 to 30 percent on

top of the existing \$0.05 to \$0.10 quality line. G4 and G9 are free because they are a comparison script and a group-by over metrics the deterministic layer already emits.

Per full golden run, 54 cases

ITEM	BASIS	PER RUN
Existing full suite	\$30-70 for 50 cases, scaled to 54	\$32 - \$76
Redaction recall (G1)	54 × ~\$0.06 Comprehend, scoring free	~\$3
Stage attribution (G2)	deterministic diff, no model	\$0
Cross-lingual end-to-end pass (C1)	54 docs × 4 languages, Haiku tier	\$5 - \$15
<i>same, Sonnet tier</i>	<i>if cross-lingual entailment needs the stronger model</i>	<i>\$15 - \$30</i>
Total		\$40 - \$110

At weekly full runs that is **\$160 to \$440 per month**, against the feasibility doc's existing \$120 to \$280 for weekly full plus deterministic-only on changes. The additions are \$40 to \$160 per month.

Why the cross-lingual pass is golden-only

This is the one recommendation that could get expensive if placed wrong. Run per production document, the end-to-end pass is roughly \$0.10 per document at the Haiku tier and \$0.28 at Sonnet, because it repeats fact extraction and matching once per language. At n=100 that is \$10 to \$28 per month, which is affordable, but it roughly doubles the per-document cost for a measurement that mostly does not vary per document.

End-to-end accuracy is a property of the pipeline configuration, not of an individual parent's IEP. It moves when a prompt, a model version, or the glossary changes, and those are exactly the events that trigger a golden run. Putting it on the golden corpus catches every real change and costs \$5 to \$15 instead of \$10 to \$28 a month. If the number later proves to drift between golden runs, promoting it to per-document is a cheap change to make.

One-time and non-dollar costs

These dominate, and they are the ones worth planning around.

ITEM	TYPE	ESTIMATE
Redaction recall harness, planted-PII ledger (G1)	engineering	~1 day
Stage attribution: per-boundary comparison (G2)	engineering	1 - 2 days
Cross-lingual end-to-end pass (C1)	engineering	1 - 2 days
Two additional G-Eval rubrics (G6)	engineering	~0.5 day
Representativeness comparison script (G4)	engineering	~0.5 day
Four extra corpus cases (G10)	generation API	~\$1
Engineering subtotal		4 - 6 days
Human calibration round: 40 summaries × 3 annotators (D1)	people	~20 hours
Native-speaker review: 4 languages × 10 docs (D1)	people	~13 hours per cycle
Independent fact-ledger audit, sample of 8 (G5)	people	~4 hours

ITEM	TYPE	ESTIMATE
Human subtotal		~37 hours per cycle

The engineering estimate sits on top of the 2.5 to 4 weeks already scoped in the feasibility doc, so the revised total is roughly 3 to 5 weeks.

What this means for the paper

Two things follow, and both belong in the methodology doc rather than only here.

First, the cheapest recommendations are the most important ones. Redaction recall costs \$3 per run and one engineering day, and it is the number that establishes the paper's entire premise. Stage attribution is free at the API and prevents every OCR failure from being misread as a summarization failure. Neither is a budget question.

Second, the binding constraint is human hours, not spend. At \$26 to \$58 recurring and \$160 to \$440 in golden runs, the whole programme costs less per month than a single day of engineering time. What actually gates it is the 37 hours of human review per calibration cycle, and that is the thing to schedule and protect. It is also worth saying plainly in the paper: under a confidentiality constraint, human judgement is the scarce input, which is precisely why so much of the design is built to run without it.

PART 6 · VERIFY BEFORE PUBLISHING

Claims I could not check from the codebase. Each is the kind of thing a reviewer checks first.

- **"the languages we are targeting consistently perform well across models (Chataigner 2024)."** That paper is about hallucination gaps *across* languages, and Vietnamese and Arabic are usually at the weaker end of multilingual evaluations. I have not read it, so I am flagging rather than contradicting. If the paper does not support the sentence, the honest version strengthens your argument anyway: Spanish is consistently strong, Vietnamese and Arabic are less well covered, which is itself the argument for the per-language human review you have already planned.
- **"The MQM Council develops and maintains the metrics."** Verify the organisational name. The framework's lineage runs through QTLaunchPad, DQF-MQM and ASTM standardisation, and the attribution should match how the maintainers describe themselves.
- **Dates.** Several entries read "Retrieved August 6, 2026" and "(2026, April 24)". Confirm these are correct rather than autofilled.
- **COMET-Kiwi licensing.** The feasibility doc records CC-BY-NC-SA-4.0 for the model weights, with non-commercial sign-off as an open item. If the published methodology names COMET-Kiwi as an alternative, a one-line note that its licence requires review is worth including.